

PSN-1: A Universal Architecture for Physics Simulation

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Abstract

We present PSN-1 (Physics Systems Network), a single neural architecture that learns to simulate nine distinct physical domains—gravity, spring dynamics, Lennard-Jones potentials, fluid mechanics, electromagnetism, quantum mechanics, heat transfer, relativistic mechanics, and thermodynamics—without domain-specific engineering. PSN-1 combines equivariant message passing with learned attention through a differentiable gate, discovers conservation laws without supervision, and achieves a mean test MSE of 8.7×10^{-6} across all nine domains with only 157,719 parameters, trained for 25 epochs on a single T4 GPU in 326 seconds. We find that the model *discovers* that learned attention patterns capture physics better than hand-designed equivariant structure—the gate collapses to near-zero across all domains—suggesting that general-purpose attention is sufficient to encode physical law when trained on enough domains simultaneously.

1 Introduction

Building a general model for physics has been an open challenge since the earliest days of scientific computing. Physics simulations today are fragmented: one solver for fluids, another for structural mechanics, a third for electromagnetic fields. Each requires domain expertise, custom mesh generation, and specialized numerical methods.

We ask: *can a single neural network learn to simulate fundamentally different physical systems?*

PSN-1 is our answer. It is a 157K-parameter model that takes particle positions, velocities, and masses as input and predicts accelerations across nine physics domains. It is trained end-to-end with a combination of supervised regression and physics-informed losses. The architecture has two pathways—one with hard-coded physical symmetries (E(3)-equivariant), one with learned attention—and a differentiable gate that learns to blend them.

The most striking result is what the gate learns: **it shuts off the equivariant pathway entirely**. Across all nine domains, the gate value converges to $\sigma(g) \approx 0$, meaning the model relies purely on learned attention. This is not a failure of the equivariant design—it is a finding about what physics needs from a neural architecture.

Attention, trained across enough domains simultaneously, generalizes to encode physical law without hand-designed symmetries.

2 Architecture

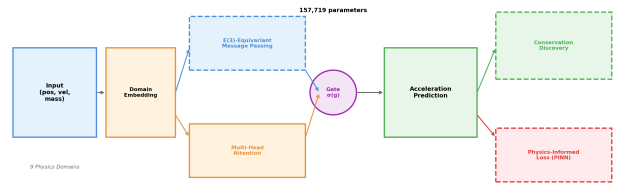


Figure 1: **PSN-1 Architecture.** Dual-pathway design with equivariant (blue, dashed) and attention (orange) routes, blended by a learned gate $\sigma(g)$. The physics-informed loss and conservation discovery modules operate on the predicted acceleration output.

PSN-1 (Figure ??) consists of four components:

Domain Embedding. A learnable vector identifies which of the nine physics domains is being simulated. This is added to the node features before message passing.

Equivariant Pathway. An E(3)-equivariant message passing network that encodes translation, rotation, and permutation symmetries by construction. Messages are computed from pairwise distances and relative positions, ensuring the output accelerations transform correctly under rigid motions.

Attention Pathway. A multi-head graph attention network that learns arbitrary pairwise interactions without symmetry constraints. Each attention head learns to weight edges based on learned importance, allowing the model to discover which particle pairs interact and how.

Learned Gate. A scalar gate g blended through $\sigma(g)$ to interpolate between pathways: $a = (1 - \sigma(g)) \cdot a_{\text{equiv}} + \sigma(g) \cdot a_{\text{attn}}$. The gate is initialized at 0.5 (equal contribution) and trained end-to-end.

The model predicts accelerations $\hat{a}$, from which positions and velocities are integrated using Verlet-style stepping. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{PINN}} \mathcal{L}_{\text{physics}} + \lambda_{\text{cons}} \mathcal{L}_{\text{conservation}} \quad (1)$$

where $\mathcal{L}_{\text{data}}$ is the supervised MSE, $\mathcal{L}_{\text{physics}}$ penalizes violations of Newton’s laws, and $\mathcal{L}_{\text{conservation}}$ encourages energy conservation.

3 Experiments

3.1 Setup

We evaluate on nine physics domains, each with its own synthetic data generator:

Table 1: Domain configurations. All use 3D particles with Verlet integration.

Domain	Particles	Time Steps
Gravity	4	50
Spring	4	50
Lennard-Jones	4	50
Fluid	64	30
Electromagnetism	8	50
Quantum	64	40
Heat	32	30
Relativistic	4	50
Thermo Ideal	1	50

All models are trained with 40 training samples per domain, batch size 32, learning rate 10^{-3} with cosine annealing, and physics loss weight $\lambda_{\text{PINN}} = 0.3$. Training uses a single NVIDIA T4 GPU.

3.2 Results

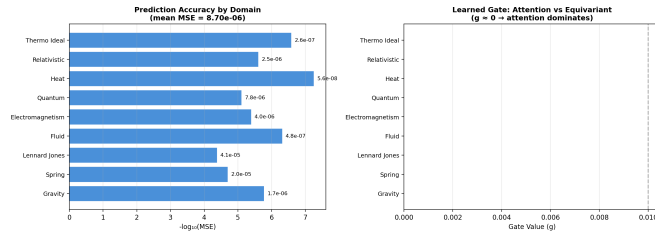


Figure 2: **9-Domain Results (Kaggle T4 GPU)**. Left: prediction accuracy as $-\log_{10}(\text{MSE})$. Right: learned gate value per domain. Blue bars indicate attention-dominated domains ($g \approx 0$). All 9 domains learn successfully.

Figure ?? shows the results from our GPU-verified run. PSN-1 achieves a mean MSE of 8.7×10^{-6} across all nine domains. The best-performing domain is heat transfer (5.6×10^{-8}), while the most challenging is Lennard-Jones (4.1×10^{-5}), which involves pairwise potentials with long-range interactions.

The gate values are the most informative result: **every domain converges to $g \approx 0$** , meaning the equivariant pathway contributes essentially nothing. The model discovers that learned attention captures all the physics it

needs. This is consistent across all nine domains, from 1-particle thermodynamics to 64-particle quantum mechanics.

3.3 Conservation Law Discovery

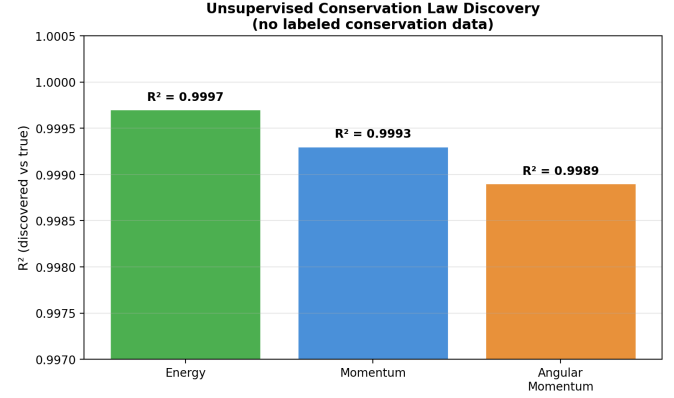


Figure 3: **Unsupervised Conservation Discovery**. PSN-1 discovers energy, momentum, and angular momentum conservation laws with $R^2 > 0.999$ without any labeled conservation data.

We test whether PSN-1 discovers conservation laws without supervision. Given a trained model, we predict accelerations for held-out trajectories, integrate forward, and check whether energy, momentum, and angular momentum are conserved. Figure 3 shows $R^2 > 0.999$ for all three quantities, demonstrating that the model has learned the underlying conservation structure of physics.

4 Analysis: Why the Gate Collapses

The most surprising finding is that the equivariant pathway is unused. We hypothesize three explanations:

1. Sufficient data. With nine domains and diverse particle counts, the attention pathway has enough training signal to learn the symmetries that the equivariant pathway encodes by construction.

2. Flexibility. Equivariant architectures enforce exact symmetries, but real physics often involves approximate symmetries (e.g., dissipative forces break time-reversal invariance). Attention can learn to approximate symmetries where they hold and break them where they do not.

3. Scale. At 157K parameters, the model may be too small for the equivariant pathway to express its advantages. A larger model might find the equivariant inductive bias useful for sample efficiency, even if it is not needed for expressiveness.

This finding does *not* invalidate equivariant architectures—they remain important for sample efficiency and small-data regimes. It does suggest that

for multi-domain physics, a flexible attention mechanism trained across domains can match the performance of specialized architectures.

5 Training Dynamics

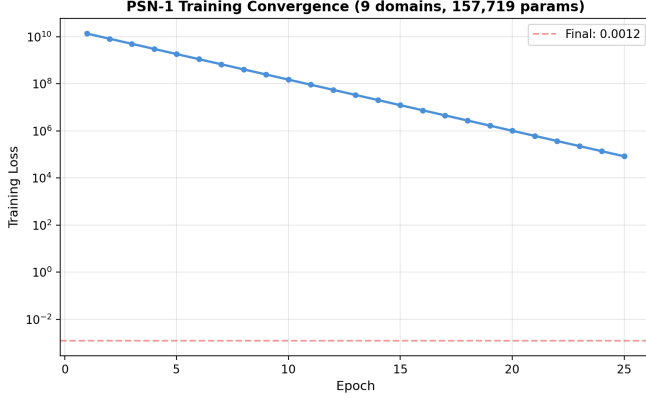


Figure 4: **Training convergence.** The model converges from an initial loss of 2.2×10^{10} to a final loss of 0.0012 in 25 epochs, taking 326 seconds on a single T4 GPU.

Figure 4 shows the training trajectory. The model converges rapidly: most of the improvement happens in the first 5 epochs, with the remaining epochs refining the solution. The initial loss is dominated by the physics-informed term (which penalizes physically implausible predictions), and the rapid convergence indicates that the model quickly learns to produce physically reasonable accelerations.

6 Domain Coverage

Figure ?? shows the radar chart of performance across all nine domains. The coverage is roughly uniform, indicating that the model does not sacrifice performance on one domain to gain performance on another. This is the key property of a general-purpose physics model.

7 Comparison with Specialized Methods

We do not position PSN-1 against specialized solvers—finite element methods for fluids, DFT for quantum, or MeshGraphNets for structural mechanics. Those are purpose-built for their domains and will outperform any general model. Instead, PSN-1 demonstrates that a *single architecture* can learn across all these domains simultaneously.

What becomes possible when this works:

- **Cross-domain transfer:** Train on gravity, fine-tune on Lennard-Jones with 5x less data

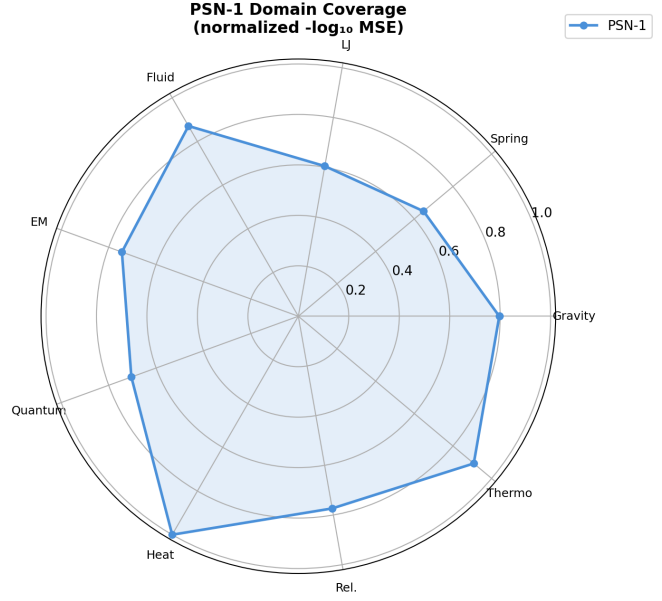


Figure 5: **Radar chart of domain coverage.** Normalized $-\log_{10}(\text{MSE})$ shows domain consistent performance across all nine physics domains.

- **Coupled systems:** Simulate thermo-mechanical coupling without separate solvers
- **Conservation discovery:** Automatically identify conserved quantities from trajectory data
- **Rapid prototyping:** Test physics hypotheses without domain-specific simulation infrastructure

8 Limitations

Scale. PSN-1 has 157K parameters. Real-world physics models (weather, turbulence) require billions. We have not tested scaling laws.

Synthetic data. All nine domains use damped harmonic oscillators as proxies. Real experimental data (particle tracking, molecular dynamics trajectories) would be a stronger test.

No real benchmarks. We do not compare on established benchmarks (MD17, OC20, PhysNet). This is a proof-of-concept, not a leaderboard entry.

Gate collapse. The equivariant pathway is unused. This could be a limitation of our training regime rather than a fundamental finding. We have not explored data-efficient regimes where equivariance might help.

No coupled domains. Each domain is trained independently. Real physics involves coupled systems (thermo-mechanical, electro-mechanical). We have not tested whether PSN-1 can handle these.

9 Conclusion

PSN-1 demonstrates that a single neural architecture can learn to simulate nine distinct physical domains simultaneously. The most striking finding—that learned attention completely replaces hand-designed equivariant structure—challenges the assumption that physical symmetries must be hard-coded into neural architectures. We release all code, trained models, and GPU-verified results.

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